

① Karnaugh-Vereinde und expressen Bestimmen

1.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

c \ AB	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$F = C + AB$

2.

A	B	C	G
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

c \ AB	00	01	11	10
0	0	1	0	0
1	1	1	1	1

$G = C + \bar{A}B$

3.

A	B	C	D	H
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

c \ AB	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	1	1
10	1	0	1	1

$H = \bar{B} + AC$

② Simplificação com álgebra booleana

$$\textcircled{6} F = AB + \cancel{A\bar{B}} + \cancel{\bar{A}B}$$

$$F = AB$$

$$\textcircled{7} F = (\cancel{A+B})(\cancel{A+\bar{B}})$$

$$F = A$$

$$\textcircled{8} F = \cancel{A\bar{B}C} + \cancel{A\bar{B}C} + \cancel{AC\bar{B}} + \cancel{AC\bar{B}}$$

$$F = AC$$

$$\textcircled{9} F = \cancel{\bar{A}\bar{B}} + \cancel{\bar{A}\bar{B}} + \cancel{A\bar{B}C}$$

$$F = \bar{A} + \bar{B}C$$

$$\textcircled{10} F = \overbrace{A + \bar{A}B}$$

$$(\cancel{A+\bar{A}})(\cancel{A+B})$$

$$C = A+B$$

$$\textcircled{11} F = (\cancel{A\bar{B}+C})(\cancel{A+\bar{B}+C})$$

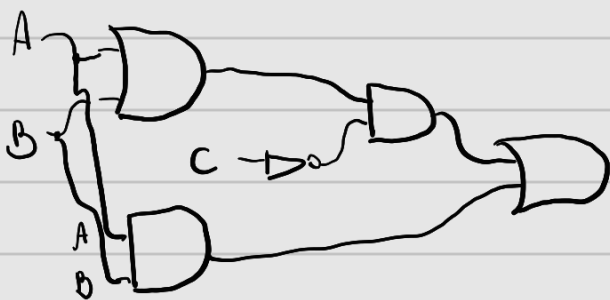
$$F = A+C$$

$$\textcircled{12} F = AB + \bar{A}C + BC$$

$$F = AB + \bar{A}C$$

③ Expressões para circuitos

$$\textcircled{13} F = (A+B)\bar{C} + AB$$



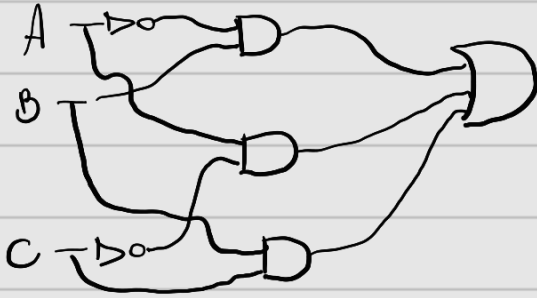
$$\textcircled{14} F = (\cancel{A+B})C + \bar{A}\bar{B}$$



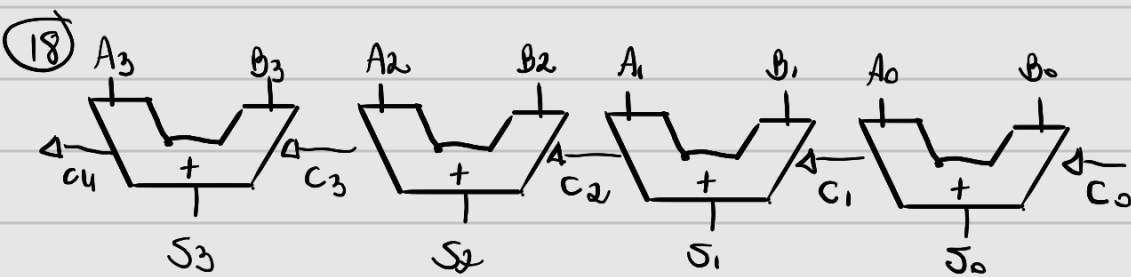
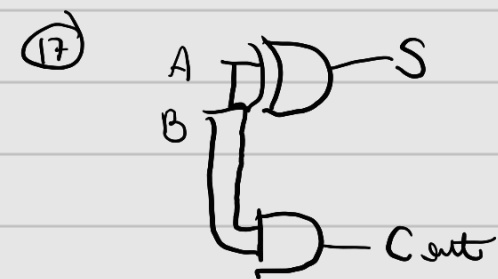
(15) $F = \overline{AB}$



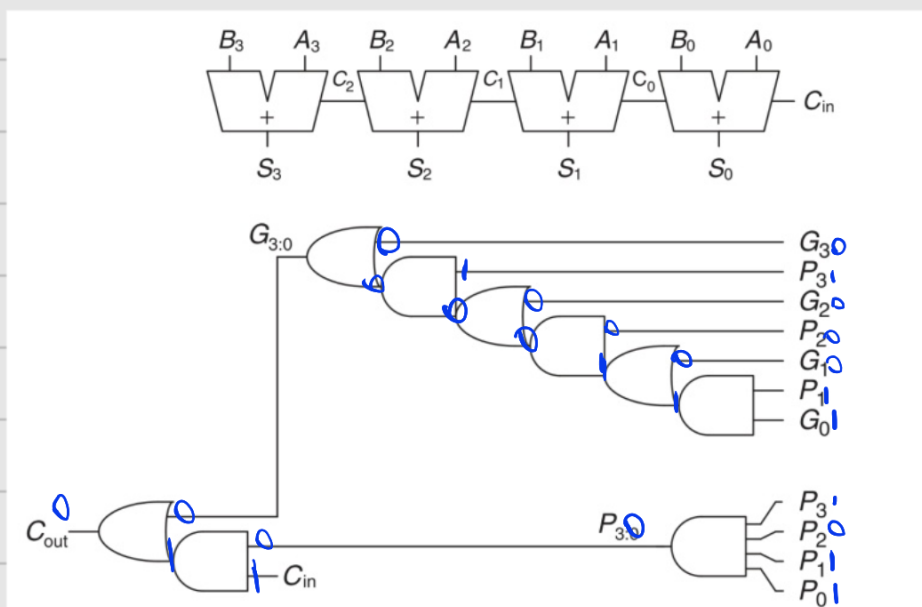
(16) $F = \overline{A}B + A\overline{C} + BC$



(14) Summary

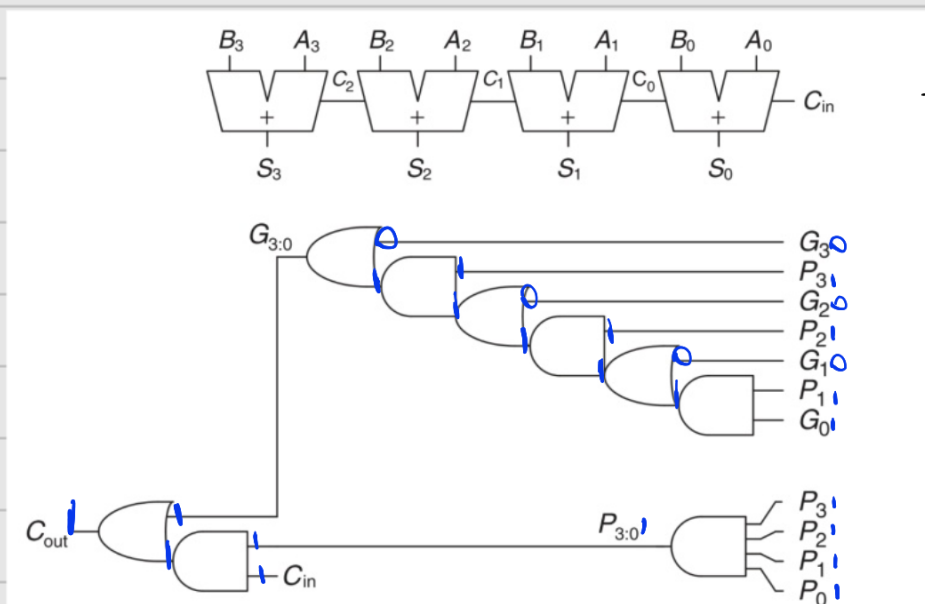


19 a



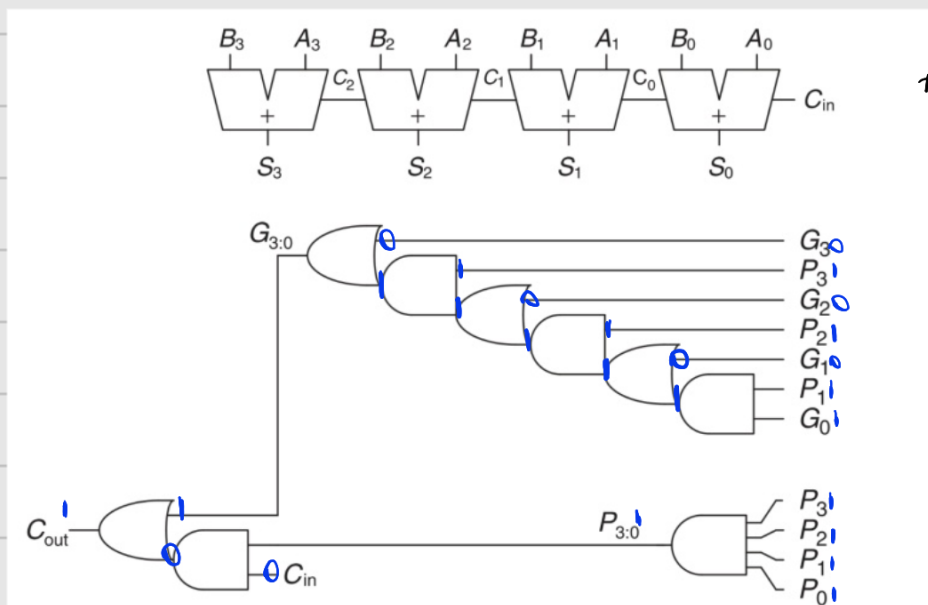
3 2 1 0
1 0 1 1
+ 0 0 0 1
C_{in} = 1

18



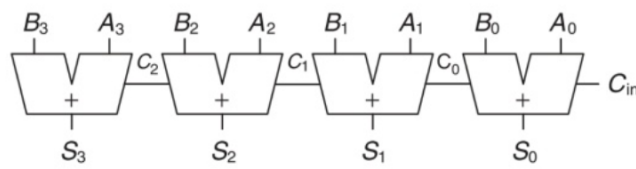
3 2 1 0
1 0 1 1
+ 0 1 0 1
C_{in} = 1

17

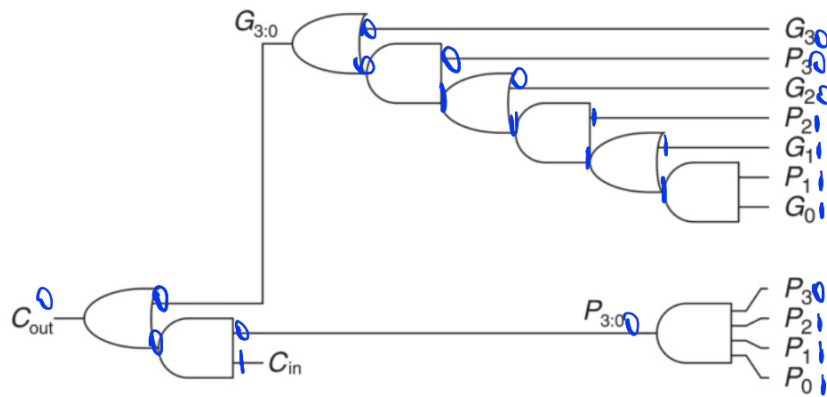


3 2 1 0
1 0 1 1
+ 0 1 0 1
C_{in} = 0

d



3 2 1 0
0 0 1 1
+ 0 1 1 1
cin = 1



5) Circuitos para expressões

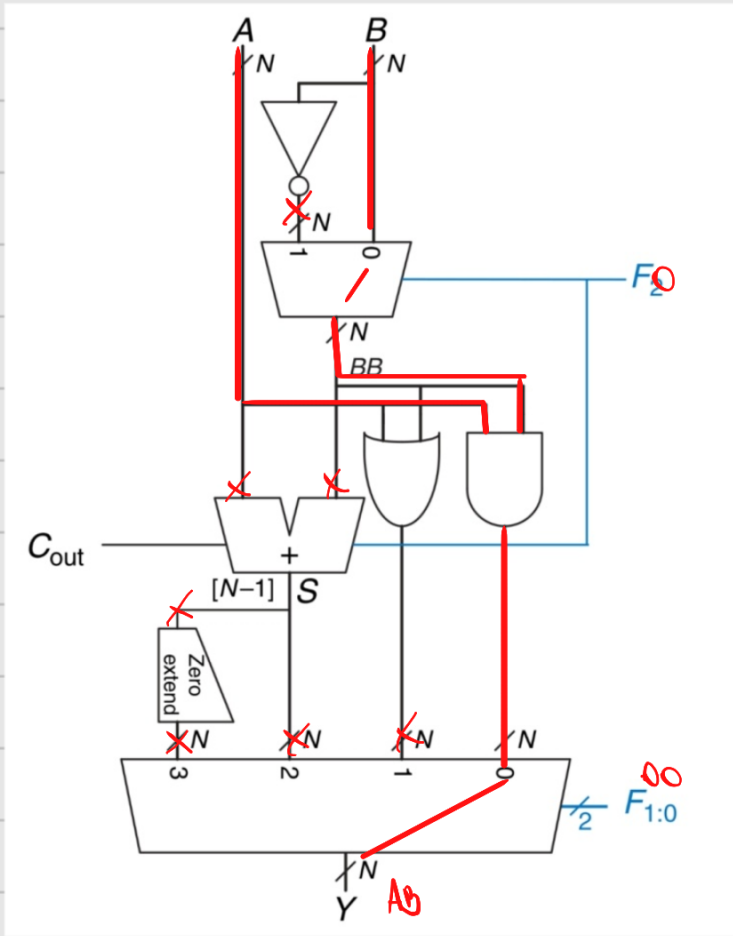
20) $Y = A\bar{B} + AB + AC$
 $Y = A$

21) $G = A\bar{C} + B\bar{C} + AC$
 $G = A + BC$

22) $X = A \oplus B$

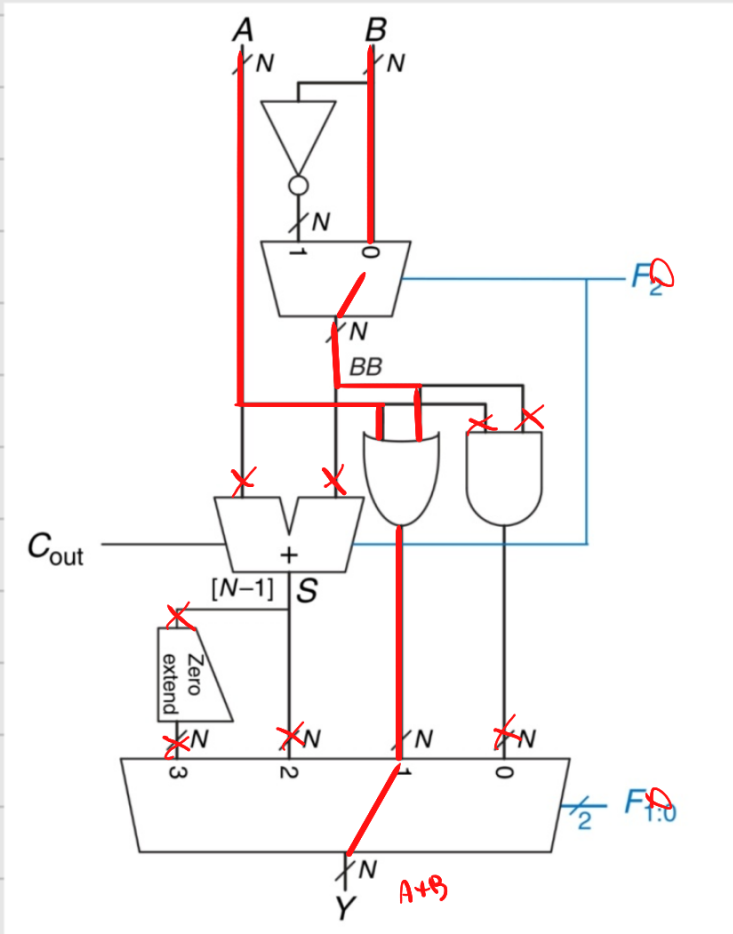
6

A AND B



$Y = AB$

A OR B



$Y = A + B$

SLT AB 

A-10

B-15

